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Tunnelling and Underground Space Technology incorporating Trenchless Technology Research

journal homepage: www.elsevier.com/locate/tust

A novel sensor network based real-time automatic longitudinal ventilation smoke control system for tunnels: A numerical investigation

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ARTICLE INFO

Keywords:

Tunnel fire
Smoke control
Ventilation
IoTs
AC
CFD
FDS

ABSTRACT

Longitudinal ventilation is widely used for smoke control in tunnel fires. This paper proposes and numerically investigates a novel ventilation scheme for tunnel smoke control that can automatically adjust its ventilation velocity in real time and in which the concepts of the Internet of Things (IoTs) and the Automatic Control (AC) are applied. Specifically, a sensor network and the PID (Proportional-Integral-Derivative) control algorithm are organically combined to output real-time transient velocity. The sensor network detects the smoke and the collected information is input into the controller, so the closed-loop control is finally formed. A numerical simulation algorithm is developed to examine the proposed smoke control method, in which the sensor network, the automatic algorithm and the Fire Dynamics Simulator (FDS) are coupled. Two types of sensors are used to form the sensor networks and their performance is compared. The simulation result shows that the smoke detector network has better control performance than the heat detector network. In addition, scenarios with different fire locations are investigated, showing that the smoke can be well controlled regardless of the studied interval distance between the fire source and the key zone that needs to be protected. Both the start-up time of the controller and the stabilization time of the whole system are proportional to the distance between the fire and the key zone. Finally, the comparison of the time-averaged quasi-steady state ventilation velocity as acquired by the proposed real-time ventilation system to experimental data as obtained for a constant ventilation-based system shows good agreement, illustrating that the simulation results are reliable.

1. Introduction

Tunnels play an essential role in the fields of transportation and resource exploration in modern society and the safety issues associated with tunnels are becoming more and more crucial. Tunnel fires, a type of frequent disaster occurring in tunnels, have been the focus of research in the community of safety science and technology (Beard and Carvel, 2012). The longitudinal ventilation system is one of the smoke exhaust systems that has been extensively used for refreshing the air in tunnels. Besides, the longitudinal ventilation system is also used for controlling the smoke generated by the tunnel fires, typically preventing it from travelling upstream the tunnel. This requires that the ventilation velocity be properly selected. Taking advantage of this, the specifications of the longitudinal ventilation system installed in the tunnel need to be rationally designed to satisfy the demand for smoke control in case of fire.

In mine tunnel fires, in order to prevent the smoke from invading other tunnel branches, the upstream spread of smoke (i.e., smoke reverse flow) needs to be stopped and completely controlled by the ventilation system. Nevertheless, too high ventilation velocity may strengthen the fire or accelerate the damage to the downstream region. Hence, the critical velocity (the minimum ventilation velocity for preventing the reverse flow of smoke) is widely investigated (He et al., 2023; Hong and Fu, 2022; Huang et al., 2020; Kennedy, 1996; Li and Ingason, 2017; Lv et al., 2023; Thomas, 1958; Weng et al., 2016; Yao et al., 2022; Yousefi et al., 2021). In addition to mine tunnels, for long tunnels without bifurcations such as those used for transportation, the focus may be to protect key zones rather than the entire upstream region, so the relationship between the longitudinal ventilation velocity and the back-layering length (the length between the fire source and the smoke front upstream) has also been extensively studied (Hong et al., 2021; Hu et al., 2008; Huang et al., 2023; Meng et al., 2018; Wang et al.,

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Received 14 June 2023; Received in revised form 9 November 2023; Accepted 12 February 2024

Available online 28 February 2024

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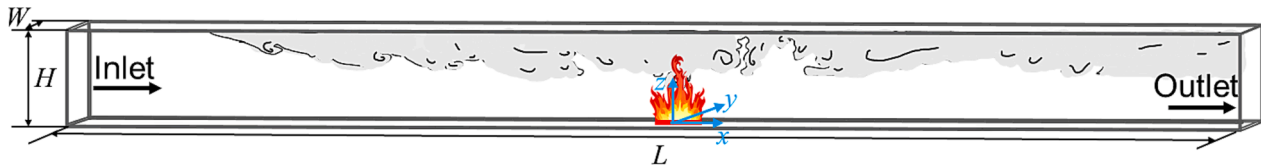


Fig. 1. Schematic diagram of a tunnel fire scenario.

2023; Yao et al., 2023; Zhang et al., 2021a). The predictive equations, correlations and machine learning models proposed in the literature are helpful to understand smoke movement under the act of ventilation systems, which is also favorable to design the specifications of the tunnel ventilation system. However, the thermal and geometric conditions may be unknown as they are not easy to be measured in real-time, so the above methods may not work to automatically adjust and control the smoke according to the site conditions in real time.

To overcome this issue, we previously proposed a novel ventilation system to automatically control the tunnel fire smoke, and numerically investigated the performance of the proposed ventilation system (Hong et al., 2022a; Hong et al., 2022b). Based on the PID (Proportional-Integral-Derivative) control algorithm, a temperature sensor is applied to record the information of the fire, and the ventilation velocity varies in real-time according to the transient and historical data of detected temperature. Though intelligent, the proposed ventilation system still has some problems: (1) the system is sensitive to the detected temperature, so the control performance may be worse for lower fire size; (2) the ventilation velocity varies too frequently with time so the system may be demanding high-quality mechanical fans; (3) To adapt to different fire sizes, the selection of the target temperature in the PID controller is not easy; (4) temperature information is currently not available easily in practice and if it were, detection would be slow and would lead to long delay times for the controller.

In this work, we aim to come up with a new novel method to overcome the above-mentioned problems. The single sensor is replaced by a sensor network (temperature sensor network and smoke detector network), and the activation state of detectors will be used as the input of the PID controller. In this way, the sensitivity of the control performance to the detected temperature can be reduced and the target temperature does not need to be estimated in advance. Besides, as the input signal in the PID controller is replaced with the number of activated sensors, the ventilation velocity does not need to change too frequently as long as the position of the smoke front does not oscillate rapidly between two sensors. Moreover, a controller startup mechanism is incorporated to reduce the delay of the controller and improve the control performance. The potential of using the proposed novel method for preventing the smoke generated by fires from spreading to the key zone in the tunnel will be numerically examined. An in-house code based on modifying the Fire Dynamics Simulator (FDS) source code is developed for that purpose and the design idea and the calculation algorithm used to implement the method are presented.

It should be noted that it is not easy to guarantee that the simulation results be completely consistent with reality since modelling the fire and the sensors is still very challenging (Schifiliti et al., 2016). Hence, it should be stressed that this paper is intended to illustrate the potential of the proposed system and that the effective elaboration of technology for practical use is considered beyond the scope of the paper.

2. Fire scenario

The configuration considered in this study is illustrated in Fig. 1: a fire in a straight horizontal tunnel with longitudinal ventilation (from left to right). The coordinate system is established as shown in the figure: the origin is at the centre of the floor, and the x , y , and z axes are parallel to the length (L), width (W), and height (H) of the tunnel, respectively. A $1 \times 1 \text{ m}^2$ square fire source occurs at the centre of the tunnel floor, with

n-heptane as unique fuel. W , H , and L of the tunnel are 3 m, 4 m, and 60 m, respectively. Although the length of real full-scale tunnels may reach several kilometers, the violently dynamic evolution of temperature and velocity fields usually occurs around the fire source. Therefore, a 60-m long tunnel with fire occurring at its center can still be used reasonably in this work. The material of the tunnel walls is assumed to be concrete with specific heat, thermal conductivity and thickness of 1.04 kJ/(kg·K), 1.8 W/(m·K) and 2 m, respectively. The velocity profile at the inlet is uniform and its magnitude is adjusted in real time.

3. Methodology

Fig. 2 shows the overview of the smoke control system. The detectors capture the fire and smoke information. Detectors are evenly installed beneath the tunnel ceiling with a specified interval. By referring to the Code for Design of Automatic Fire Alarm System (China, 2008), a 2-m interval is chosen in this study. Every detector has a signal transmitter to send its state: 1 if it is activated, 0 otherwise. It should be clarified that in practice, for convenience and economy, the line-type heat detector is usually used to detect tunnel fires instead of point-type heat detector. However, this work aims to use the activation information provided by detectors/sensors to control tunnel fire smoke, so point-type sensors are used. Although point-type detector networks do not have economic advantages, this cost is very low compared to the total cost of tunnel engineering if the system is helpful in improving tunnel safety. Hence it is believed that when the benefits of the proposed system are appreciated, there will be willingness to make this relatively cheap investment in point detectors.

This study focuses on the protection of the key zone in the tunnel and the fire is assumed to occur downstream of the key zone. The situation where the fire occurs directly in the key zone is not considered (more urgent rescue and/or fire suppression measures should be initiated in this case). To implement this goal, the tunnel is nominally divided into four parts:

- (I) **Key zone**, which needs to be protected.
- (II) **Tolerable zone**, which, as a buffer zone, admits the smoke reach here. This zone is defined as the region that includes the two closest sensors near the boundary (right side) of the key zone.
- (III) **Free zone**. This zone is defined as the area from the fire source to the (right side) boundary of the tolerable zone. The smoke can stay here freely. In this zone, the detectors can freely send their activation status signal once activated.
- (IV) **Other zones**. The downstream region in the tunnel. The detectors in this zone are not permitted to send their activation status signals.

Overall, the detectors in Zone (I)-(III) are permitted to send their activation status signal freely, but they are not permitted to send their activation status signal in Zone IV.

Once the fire occurs, the smoke generated will rise vertically. In this study, the ventilation system is default set as 'turn off' if no sensor is activated. On this premise, therefore, the upstream and downstream can be determined from the location of the first activated sensor: the location of the first triggered sensor to the outlet of the tunnel is considered the downstream region. After that, the four zones can be determined immediately, and the downstream detectors no longer send their

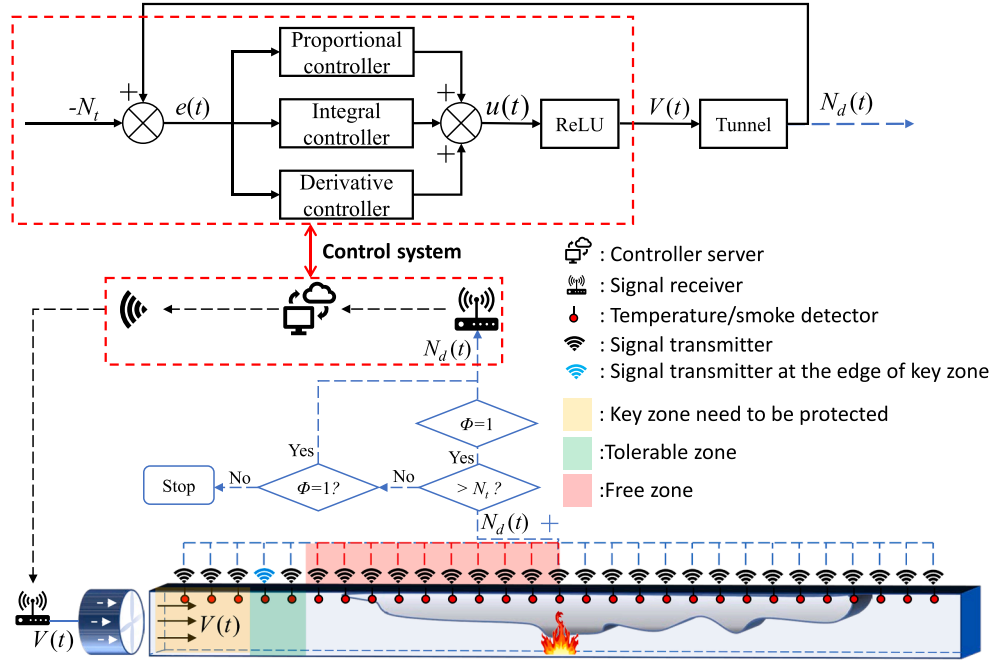


Fig. 2. Schematic diagram of smoke control in a tunnel fire using automatic control algorithms and the IoT.

activation status signals. The identification of the downstream and upstream regions can also estimate the location of the fire source, i.e., the fire source is around the first activated sensor within a distance equal to half the distance between two adjacent sensors (in the absence of tunnel slope). It is noted that this judgment is based on the premise that the ventilation system is in the status ‘turned on’ before the fire occurs. If the ventilation system is ‘turned off’ before the fire occurs, the smoke may travel some distance before the smoke activates the first sensor. In this case, the traveling distance of the smoke needs to be considered, but this complex case will not be discussed in this work. Moreover, the tunnel ceiling height is typically low, so this distance should be limited. An identifier, Γ , marks whether the fire source is found: Γ initially equals 0 and becomes 1 after one sensor is activated. Once Γ equals 1, the process of identifying the fire source location ends in order to save computing resources.

The sensors in the upstream zone feed into the control system: their activation statuses are the critical basis for adjusting the ventilation velocity. The number of activated sensors, marked as $N_d(t)$, is updated in real time (e.g., on a cloud server). When the location of fire source and the key zone are known, the number of sensors in the free zone can be inferred and calculated automatically. The number of sensors in the free zone is marked as N_t ; it is noted that this value is fixed once the first sensor is activated. The goal of the ventilation system is to control the smoke and keep it away from the key zone. Therefore, the system should make $N_d(t)$ approach N_t in the control process: $N_d(t) \rightarrow N_t$. This will correspond to the lowest ventilation velocity meeting the target of smoke control at the correct distance.

The PID control algorithm is adopted to achieve this goal. The difference between $N_d(t)$ and N_t defines the error function ($e(t) = N_d(t) - N_t$) and serves as an input to the control algorithm. The ventilation velocity ($u(t)$) serves as an output of the PID controller. The algorithm can be written as

$$u(t) = K_p e(t) + K_i \int_0^t e(t) dt + K_d \frac{de(t)}{dt}, \quad (1)$$

where K_p is the coefficient of the proportional controller, K_i is the coefficient of the integral controller and K_d is the coefficient of the derivative controller. These parameters need to be tuned for obtaining

good performance and will be discussed below. Through this algorithm, $e(t)$ can converge to zero with the dynamic adjustment of the ventilation velocity. It is noted that the ventilation direction is one-way, so the ventilation velocity is modified by a ReLU function:

$$V(t) = \max\{u(t), 0\}. \quad (2)$$

A closed loop control is formed, as shown in Fig. 2. This control algorithm can be assembled in the cloud server, so it can run as long as it receives activation signals from the signal transmitter.

In addition, as $e(t)$ remains negative while the smoke travels to the tolerable zone, the term $K_i \int_0^t e(t) dt$ would accumulate the negative error function. As a result, the time for $u(t)$ from negative to positive becomes so long that the smoke may have travelled deep into the key zone. This would lead to too much delay for launching the ventilation system, because the ventilation system turns on only if $u(t) > 0$. To solve this delay issue, we let the PID controller start to work only when $N_d(t) \geq N_t$ and remains activated once it has started working. This process is shown in the program diagram in Fig. 2, where Φ is an identifier, which is initially 0 and remains 1 once:

$$N_d(t) \geq N_t. \quad (3)$$

The above process can be easily executed on the cloud server as well.

It must be stressed that in practice, $N_d(t)$ can only be expected to be monotonically (and step-wise) increasing with state-of-the-art point detectors, where normally ‘old’ smoke is not purged. Hence, once activated, it cannot be expected for the detector to de-activate (whereas this is an implicit assumption in the methodology as adopted here). In order to make that possible, smoke would need to be purged (e.g., through aspiration), but this would introduce additional time constants, which could be a challenge for the steering of the system. These practical aspects are considered important, but are left for future work at this stage.

4. Numerical models and solutions

4.1. Models and settings in CFD simulation

The simulations are performed with an in-house code, combined

with the source code of Fire dynamics simulator (FDS) version 6.7.7 (McGrattan et al., 2021): the CFD is done by FDS while the automatic control algorithm and the interactive logic of the sensor networks are assembled in the source code. In the simulation, the uniform velocity at the inlet of tunnel is updated according to Equation (1) in real time. The outlet of tunnel is given the 'OPEN' boundary condition (McGrattan et al., 2021). The yields of soot and CO are specified as 0.037 kg/kg and 0.01 kg/kg for the n-heptane fuel, respectively, and the radiative fraction is set to 0.4 (Beyler, 2016). The ambient temperature is 24 °C. The settings of material, fire source and tunnel geometry are consistent with the description in section 2. Other settings follow the default values or mode in FDS version 6.7.7. The simulation time ($t_{\max}$) is 350 s, which is long enough for the system to reach the quasi-steady state condition.

The characteristic fire diameter D^* is used to obtain a first estimate for the mesh cell size:

$$D^* = \left(\frac{\dot{Q}}{\rho_{\infty} c_p T_{\infty} \sqrt{g}} \right)^{\frac{2}{3}}, \quad (4)$$

where $\dot{Q}$, ρ_{∞} , c_p , T_{∞} , g are the total HRR (heat release rate) of the fire, the ambient air density, the specific heat of air, the ambient air temperature and the acceleration of gravity, respectively. A mesh cell size of $D^*/10$ gives an initial criterion (McGrattan et al., 2013). A cubic cell of size (Δ) equal to 0.0625 m has been used in this work, satisfying the $D^*/\Delta > 10$ criterion for all calculations.

4.2. Modelling and settings of sensors

Two types of sensors, i.e., heat and smoke detectors, are tested and compared for the sensor networks in this work. For the simulation of heat detector, the sensors are considered in activated state if the detected temperature is higher than

$$T_{act} = T_{\infty} + T_{diff}, \quad (5)$$

where T_{act} and T_{diff} are respectively the activation temperature and temperature difference between the ambient temperature and the activation temperature. T_{∞} and T_{diff} are respectively set as 24 °C and 70 °C, so the activation temperature is 94 °C in this work. T_{diff} can be adjusted in practice, of course. The heat detector is essentially a thermocouple, equipped with an activation function. The detected temperature follows the thermocouple equation (McGrattan et al., 2021), with default thermocouple parameters of FDS (McGrattan et al., 2013). The activation state $n_i(t)$ reads:

$$n_i(t) = \begin{cases} 0, & T_d(t) < T_{act} \\ 1, & T_d(t) \geq T_{act} \end{cases}, \quad (6)$$

where $n_i(t)$ represents the activation state of the i -th sensor (upstream) at time t , $T_d(t)$ is the temperature detected by the sensor at time t . As a result, $N_d(t) = \sum_i n_i(t)$.

For the simulations with smoke detector, the sensors are activated according to the chamber obscuration. SFPE (Schifiliti et al., 2016) reported that the smoke detector can respond when the chamber obscuration is over 29.22 %/m in a kerosene fire. For convenience, this work

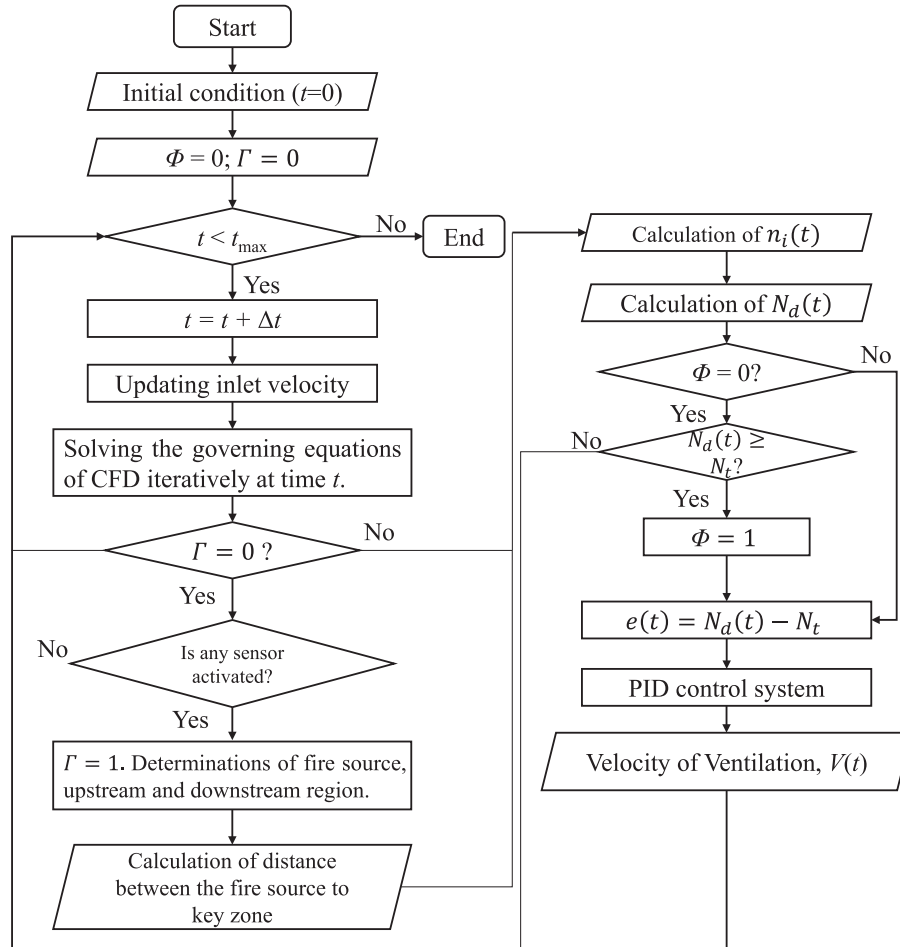


Fig. 3. Flowchart of the simulation. Δt is the time increment.

refers to this value, so the sensors are activated when

$$O_{act} > 29.22 \frac{\%}{m}, \quad (7)$$

where O_{act} is the activation obscuration. The obscuration is defined as

$$\text{Obscuration} = (1 - e^{-K_m \rho Y_c}) \times 100\%, \quad (8)$$

where Y_c is the mass fraction of smoke in the sensing chamber, K_m is the mass extinction coefficient, ρ is the density of the external gases in the ceiling jet. For most flaming fuels, a suggested value for K_m is $8700 \text{ m}^2/\text{kg} \pm 1100 \text{ m}^2/\text{kg}$ at a wavelength of 633 nm (Mulholland and Croarkin, 2000).

4.3. Solution procedure

Fig. 3 illustrates the flowchart of the solution procedure. Through logical judgments and combining the above-mentioned methods, the fire source position can be found and the number of upstream activated sensors can be calculated. Finally, the control algorithm is coupled with the CFD calculation.

5. Results and discussion

5.1. Determination of parameters

Obviously the parameters in Eq. (1) have a significant effect on the smoke control performance. Their values have been determined using the order of magnitude and trial-and-error method as previously proposed (Hong et al., 2023). K_d is set to zero because the derivative controller is sensitive to the turbulent system (Hong et al., 2023).

The order of magnitude of K_p is 1 m/s : typical ventilation velocities are in the order of (1 m/s) and the controlled smoke front oscillates within in a small zone when the difference between $N_d(t)$ and N_t is not large (so that also $e(t)$ is in the order of (1)). As a result, indeed:

$$K_p = \mathcal{O}\left(\frac{V}{e(t)}\right) = \mathcal{O}\left(\frac{1 \frac{\text{m}}{\text{s}}}{1}\right) = \left(1 \frac{\text{m}}{\text{s}}\right). \quad (9)$$

Four values have been tested for K_p : 0.5 m/s , 1 m/s , 2 m/s and 3 m/s . Too many oscillations were observed with the latter two values (see Fig. 4), so these were ruled out. The advantage of $K_p = 0.5 \text{ m/s}$ (over $K_p = 1 \text{ m/s}$) is that proportional steps in ventilation velocity are only half as large (see Eq. (1)), so that choice was made here.

A similar procedure was followed to determine K_i . The rise time of the ventilation velocity in the early control period is about 10 s (see below). Hence, the integration time is in the order of $\mathcal{O}(10 \text{ s})$. If the integral controller is added, it will work together with the proportional controller in the early control period. As a result, K_i is in the order of:

$$K_i = \mathcal{O}\left(\frac{V}{\int_0^{10} e(t)}\right) = \mathcal{O}\left(\frac{1 \frac{\text{m}}{\text{s}}}{10 \text{ s} \times 1}\right) = \mathcal{O}\left(10^{-1} \frac{\text{m}}{\text{s}^2}\right). \quad (10)$$

Four values have been tested again: (a) $K_i = 0.05 \text{ m/s}^2$, (b) $K_i = 0.1 \text{ m/s}^2$, (c) $K_i = 0.2 \text{ m/s}^2$, and (d) $K_i = 0.4 \text{ m/s}^2$. The results are shown in Fig. 5, which reveal that the lower values lead to too much overshoot in $N_d(t)$.

In Fig. 5 (d), the key zone is 10 m away from the fire source (i.e., N_t equals 4 , corresponding to a back-layering length of $3 \times 2 = 6 \text{ m}$); the HRR is 2500 kW ; the detectors are chosen as heat detectors and it is assumed that one of the detectors is installed directly above the centre of the fire source. Because the interval of the detectors is 2 m , $N_d(t) = 5$ means that the smoke back-layering length is in the range between $4 \times 2 = 8 \text{ m}$ to $5 \times 2 = 10 \text{ m}$. It is worth noting that the ventilation velocity from Eq. (11) below, as proposed by Li et al. (2010) for fixed ventilation velocity is: 2.14 m/s for back-layering length of 0 m , 1.98 m/s for back-

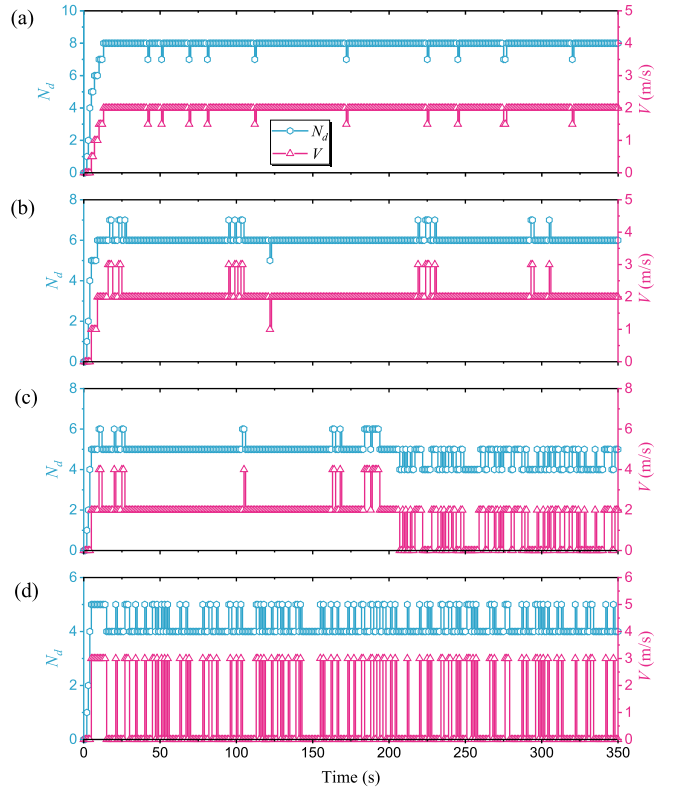


Fig. 4. The evolutions of the $N_d(t)$ and $V(t)$ under (a) $K_p = 0.5 \text{ m/s}$, (b) $K_p = 1 \text{ m/s}$, (c) $K_p = 2 \text{ m/s}$, and (d) $K_p = 3 \text{ m/s}$. K_i and K_d are set as zero.

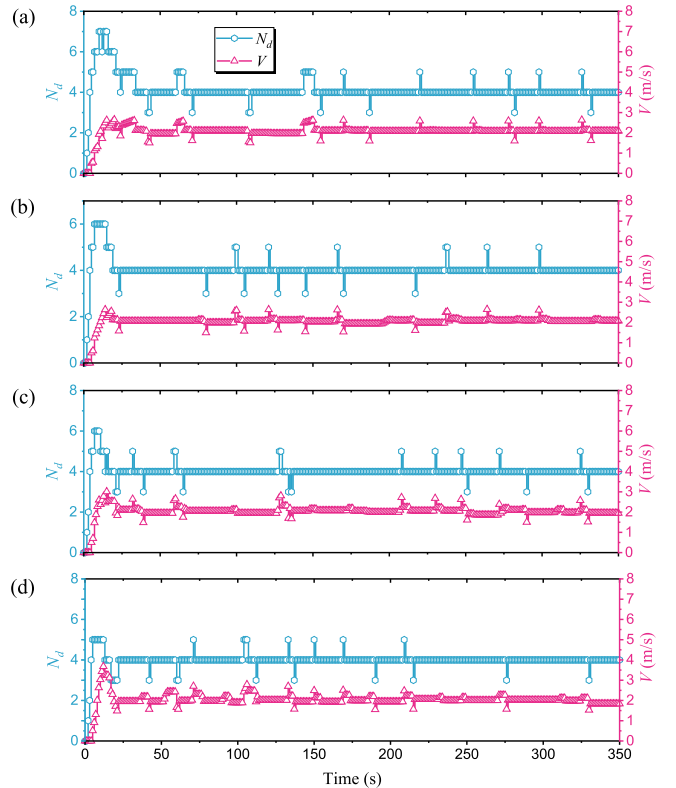


Fig. 5. The evolutions of the $N_d(t)$ and $V(t)$ respectively under (a) $K_i = 0.05 \text{ m/s}^2$, (b) $K_i = 0.1 \text{ m/s}^2$, (c) $K_i = 0.2 \text{ m/s}^2$, and (d) $K_i = 0.4 \text{ m/s}^2$. K_p and K_d are respectively set as 0.5 m/s and zero.

layering length of 6 m, and 1.77 m/s for back-layering length of 10 m. This demonstrates already the correct order of magnitude for the ventilation velocity as obtained here.

5.2. Heat detector network vs. Smoke detector network

It is noted that the smoke detectors are rarely used in practice since false alarms can be possibly caused by the exhaust emissions from vehicles. However, this section aims to discuss the performance of the two types of detectors for the proposed system, thus such false alarms are ignored. HRRs of 500 kW ~ 5 MW are respectively investigated for both types of sensors, as listed in Table 1. The key zone is set at 10 m from the fire source, similar to the settings in section 5.1.

The time-averaged (except for smoke contours) results of the temperature and soot fraction distributed at 5 cm below the tunnel ceiling along the x-direction are illustrated in Fig. 6. It can be seen that only the smoke detector network can work well when the HRR is lower than 1000 kW: the activation temperature $T_{act} = 94$ °C is a relatively high value for the HRR lower than 1000 kW, so the heat detector cannot be easily activated even when the smoke has already passed the sensor. Fig. 7 shows the transient temperature contours by using the heat detector network at different moments for the 1000 kW fire. The result confirms that the smoke cannot be controlled well during the quasi-steady period. However, since O_{act} (Eq. (7)) only varies in a small range for both small and large fires and is not as sensitive as temperature, the smoke detector network can control the smoke well even if the fire is small. Overall, the smoke detector network can work in a wider range of HRR for detecting smoke than the heat detector network.

Nevertheless, both types of sensor networks work effectively for fires with higher HRR, as shown in Fig. 6 (c1)-(e1) and (c2)-(e2). As an example, Fig. 8 shows the transient temperature contours of a 3000 kW fire at different moments by using the heat detector network: the smoke can be controlled well in zone II (tolerable zone). Besides, it is found that the temperature and soot fraction distributions under the same fire size are similar (not exactly the same), regardless of the types of detector networks. The reason why the distribution is not exactly the same is due to turbulence disturbance: the detector may not be in the state of activation at some (very rare) time even if the smoke shroud around the sensor, and the timing at which different types of detectors do not respond to smoke may vary.

From the results depicted in Fig. 6 (c1)-(e1) and (c2)-(e2), it can be found that although the shapes of temperature and soot fraction distributions are not similar, the locations where the curves rise sharply almost overlap. This reflects simultaneous transport of heat and soot of the smoke since the locations with sudden rise of the curves correspond to the location of the smoke front (Hong et al., 2022a). The grey regions in Fig. 6 are the tolerable zone, in which the left and right sides are respectively the key zone and the free zone. It is seen that the smoke fronts stay in the tolerable zone in all cases where the smoke is controlled. This result is consistent with the initial expectations: the key zone is set 10 m away from the fire source, so N_t is 4, and the tolerable zone is thus 6 m away from the fire source.

Table 1

List of the fire conditions with different types of detectors.

Case No.	HRR (kW)	Types of detectors
a1	500	Heat
a2	500	Smoke
b1	500	Heat
b2	1000	Smoke
c1	1000	Heat
c2	2000	Smoke
d1	2000	Heat
d2	3000	Smoke
e1	5000	Heat
e2	5000	Smoke

Overall, the smoke detector network has achieved better control performance because it is effective for small fires. However, the temperature sensor network is still effective for larger fires, so it still has a promising future because of its robustness (i.e., less prone to false alarms than smoke detectors). If the fire is very small, using the smoke detector network may also not work. Fortunately, small fires have already left enough time for firefighters, and our system is more effective for larger fires, which may increase the success rate of fire rescue.

5.3. Performance for different fire locations

In this section, only the smoke detector network will be considered, with the parameters of the PID controller as set in Section 5.1. The fire size is assumed as 3000 kW. This section aims to investigate the control performance for variable locations of the fire source. To simplify the simulation, the fire source is fixed at the centre of the tunnel floor, but the key zone is changed to different locations in the upstream region of the tunnel. Distances of 6 m, 10 m, 14 m, 18 m, 22 m, and 26 m between the fire source and the key zone are investigated, respectively. These cases correspond to N_t equal to 2, 4, 6, 8, 10, and 12, respectively (see Table 2).

Fig. 9 shows the evolutions of $N_d(t)$ for different distances between the fire source and the key zone. After the PID controller launches for the first time when $N_d(t) = N_t$, $N_d(t)$ will first become slightly higher than N_t to increase the ventilation velocity. After the ventilation velocity is increased to a high enough value, $N_d(t)$ decreases and is controlled to the desired values. However, it is very unlikely that the ventilation velocity is such that the air momentum equals the smoke momentum exactly. If the ventilation velocity is slightly too low, the smoke will slowly move further upstream, until it is detected by the next upstream detector, after which the velocity is increased and the smoke is pushed backward. Yet, the ventilation velocity then decreases again. Therefore, if it decreases again to a value that is slightly too low again, this cycle will repeat itself (see Fig. 10, a few positive spikes are shown such as the 57-th second). If, on the other hand, the velocity relaxes to a value that is slightly too high (in relation to the smoke momentum), the smoke is slowly pushed backward, until it deactivates one detector, after which the velocity is reduced (see the negative spikes in Fig. 10, e.g. 77-th second). After that, the velocity increases again. If it increases too much, the cycle repeats itself. If it does not increase enough, there is an evolution towards a positive spike again. In our previous work (Hong et al., 2022a), this phenomenon of spikes in $e(t)$ was not observed because temperature at a given location was used as independent variable. As temperature is a continuous variable, the ventilation velocity evolved until a certain steady smoke location was obtained (notwithstanding turbulence), determined by the set variable. Here, the independent variable is discrete, namely the location of the latest (de-)activated detector. Hence $e(t)$ increases (or decreases) stepwise, rather than continuous.

Fig. 11 displays the evolutions of the smoke front for different distances between the fire source and the key zone. The method for identifying the smoke front is the same as in Hong et al. (2022a). Similar to the evolution of $N_d(t)$, the smoke front gets into the key zone first, and then is pushed back to the position 2 m away from the key region. Taking the case of $N_t = 2$ as an example, the smoke front oscillates around the position of $x = -4$ m rather than $x = -2$ m (the desired position). This is because $N_d(t)$ equals 2 when the smoke front is in the region of $-4\text{ m} \leq x \leq -2\text{ m}$, while $N_d(t)$ equals 3 when the smoke front is in the region of $-6\text{ m} \leq x \leq -4\text{ m}$. Because the smoke front inherently trends to spread upstream, the smoke front waves around the position of $x = -4$ m, corresponding to $N_t = 3$. For the other cases, the reason why the smoke front oscillates around the position 2 m away from the key zone is the same. However, this is not problematic as the smoke front still falls in the tolerable zone rather than the key zone.

In addition, Fig. 11 shows that the smoke front curves overlap in the early stage ($N_d(t)$ less than N_t). This is because ventilation is not launched during this stage, so the movement processes of the smoke

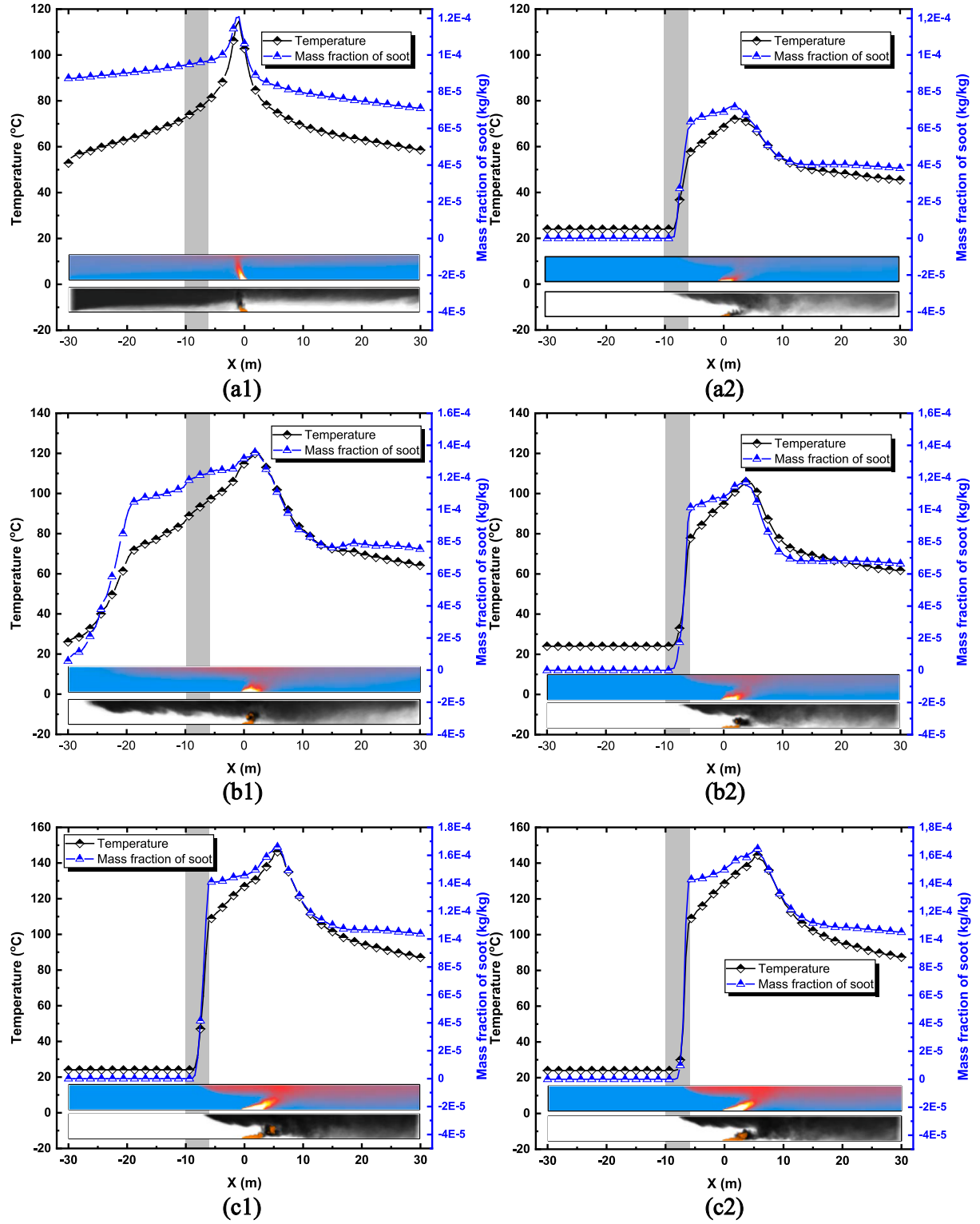


Fig. 6. The temperature and soot fraction distributions 5 cm below the tunnel ceiling along the x-axis direction, as well as the temperature and smoke contours at plane $y = 0$. (a1)-(e1) are the results by using the heat detectors, with the HRRs of 500 kW, 1000 kW, 2000 kW, 3000 kW, 5000 kW, respectively. (a2)-(e2) are the results by using the smoke detectors, with the HRRs of 500 kW, 1000 kW, 2000 kW, 3000 kW, 5000 kW, respectively.

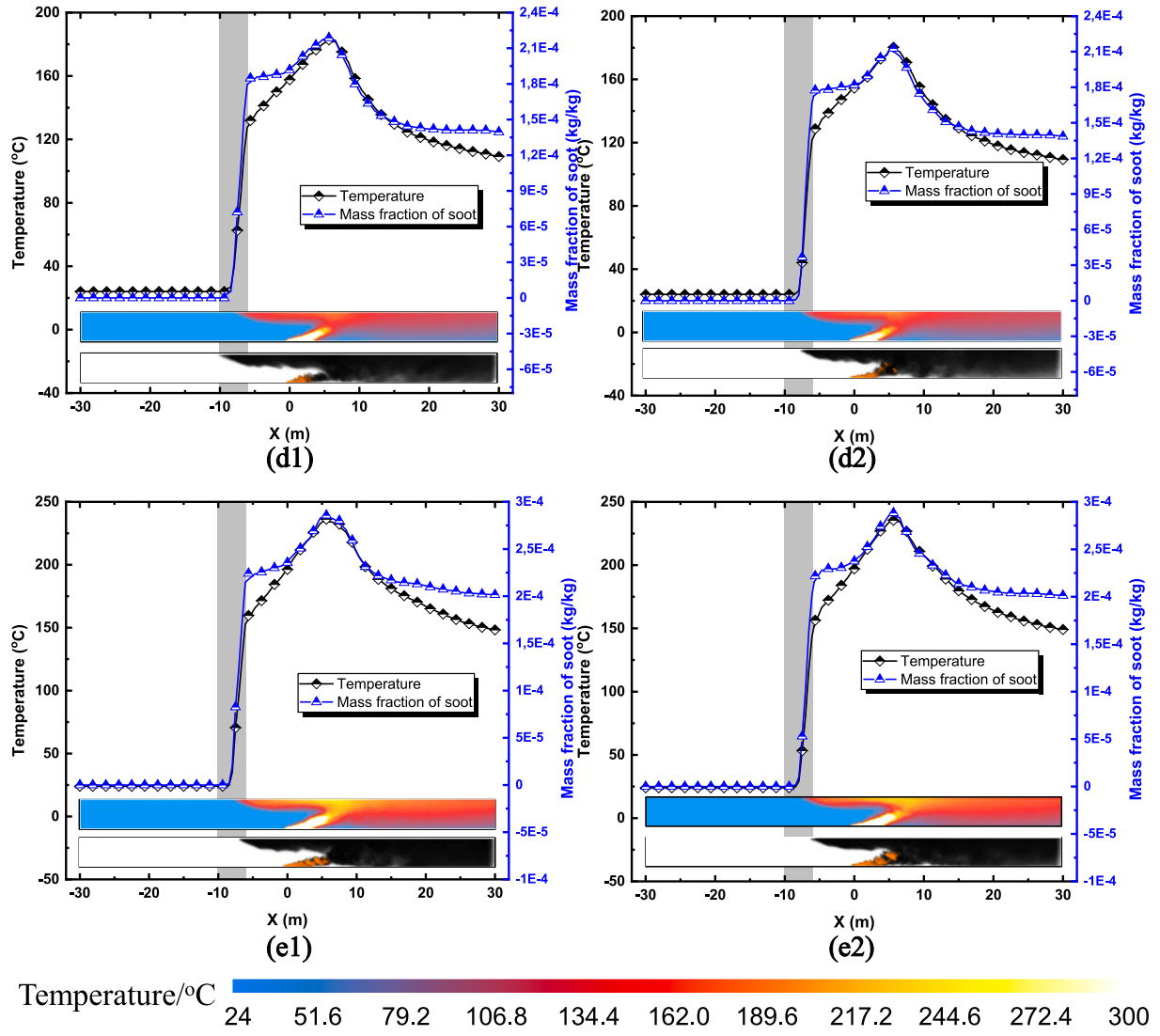


Fig. 6. (continued).

fronts in this stage are the same. Moreover, before the startup of ventilation, the smoke front moves (nearly) linearly with time. The slope is about 1.8, indicating that the velocity of smoke movement is approximately this value. Here, we define the stabilization time (t_{stb}) as the required time for the smoke front to be pushed back the first time to the position 2 m away from the desired position. Fig. 12 illustrates the relationship between the stabilization time and N_t . A (nearly) linear relationship can also be seen, which means the time between the startup of ventilation and the smoke front being pushed back to the position 2 m away from the desired position is also proportional to N_t , also shown by the blue dashed line in Fig. 11.

Fig. 13 exhibits the time-averaged temperature distributions 5 cm beneath the tunnel ceiling for different values of N_t . It is seen that the positions for sudden rise of the temperature curves are approximately at $x = -4$ m, $x = -8$ m, $x = -12$ m, $x = -16$ m, $x = -20$ m, and $x = -24$ m for $N_t = 2$, $N_t = 4$, $N_t = 6$, $N_t = 8$, $N_t = 10$, and $N_t = 12$, respectively.

This result confirms again that the proposed system has good control performance since the smoke front falls in the tolerable zone in each case. For higher N_t , the back-layering length is also larger; the ventilation velocity is thus lower. As a result, the temperature is overall higher for higher N_t .

5.4. Comparison with the experimental (fix-velocity ventilation system) result

In order to validate the accuracy and the reliability of the simulation results, this section compares the simulation results with experimental data. The experimental investigations about the longitudinal ventilation for tunnel fire smoke control found in the literature have all been achieved with constant ventilation velocity (Hong and Fu, 2022; Jiang et al., 2022; Li et al., 2010; Zhang et al., 2021b). Therefore, as far as the authors are aware, no experimental studies or correlations can be utilized

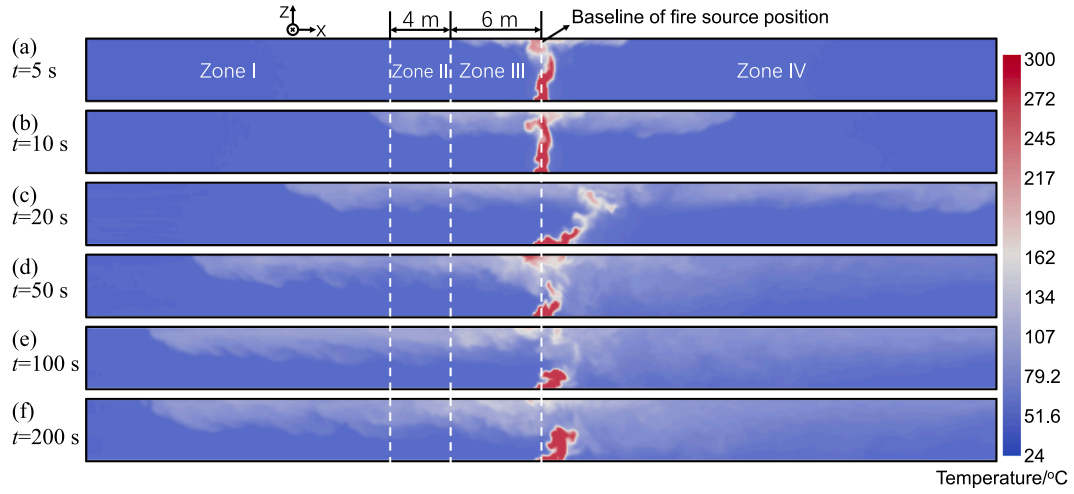


Fig. 7. Instantaneous temperature contours for the 1000 kW fire by using the heat detector networks at (a) 5 s, (b) 10 s, (c) 20 s, (d) 50 s, (e) 100 s, and (f) 200 s, respectively.

directly to validate our transient results. However, the validation method in (Hong et al., 2022b) can be imitated: taking the system after the movement of the smoke front reaches the quasi-steady state, the

time-averaged ventilation velocity in the quasi-steady state for a specific back-layering length can be compared with the constant ventilation velocity for the same back-layering length under the fixed ventilation

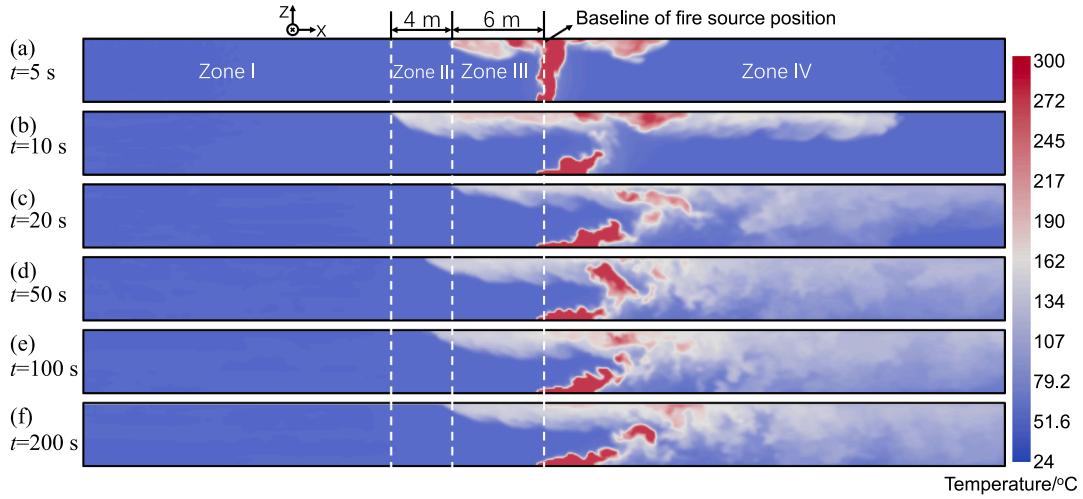


Fig. 8. Instantaneous temperature contours for the 3000 kW fire by using the heat detector networks at (a) 5 s, (b) 10 s, (c) 20 s, (d) 50 s, (e) 100 s, and (f) 200 s, respectively.

Table 2

List of the fire cases for different distances between the fire source and the key zone.

Case No.	Distances (m)	N_t
1	6	2
2	10	4
3	14	6
4	18	8
5	22	10
6	26	12

velocity system .

According to the correlation proposed by Li et al. (2010), for any known back-layering length d , the corresponding ventilation velocity produced by the constant ventilation system can be written as

$$\tilde{V} = \begin{cases} 0.81 \sqrt{gH} \left(\frac{\dot{Q}}{\rho_{\infty} c_p T_{\infty} g^{\frac{1}{2}} H^{\frac{5}{2}}} \right)^{\frac{1}{3}} e^{-\frac{d}{18.5H}}, & \frac{\dot{Q}}{\rho_{\infty} c_p T_{\infty} g^{\frac{1}{2}} H^{\frac{5}{2}}} \leq 0.15 \\ 0.43 e^{-\frac{d}{18.5H}}, & \frac{\dot{Q}}{\rho_{\infty} c_p T_{\infty} g^{\frac{1}{2}} H^{\frac{5}{2}}} > 0.15 \end{cases} \quad (11)$$

Fig. 14 shows the evolutions of the ventilation velocity and $N_d(t)$ in this study, under the HRR of 3000 kW and the back-layering length of

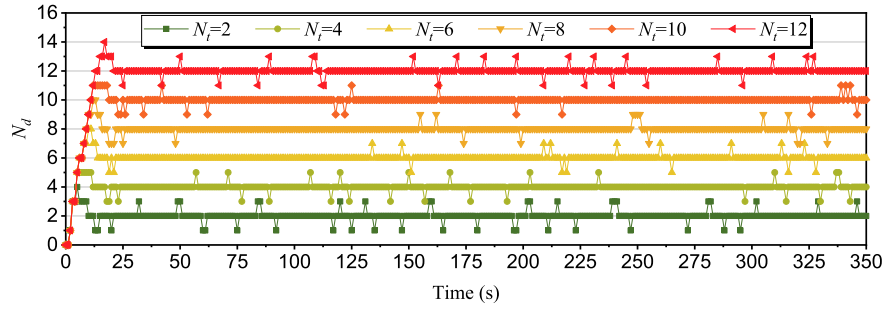


Fig. 9. Evolutions of $N_d(t)$ for different distances between the fire source and the key zone.

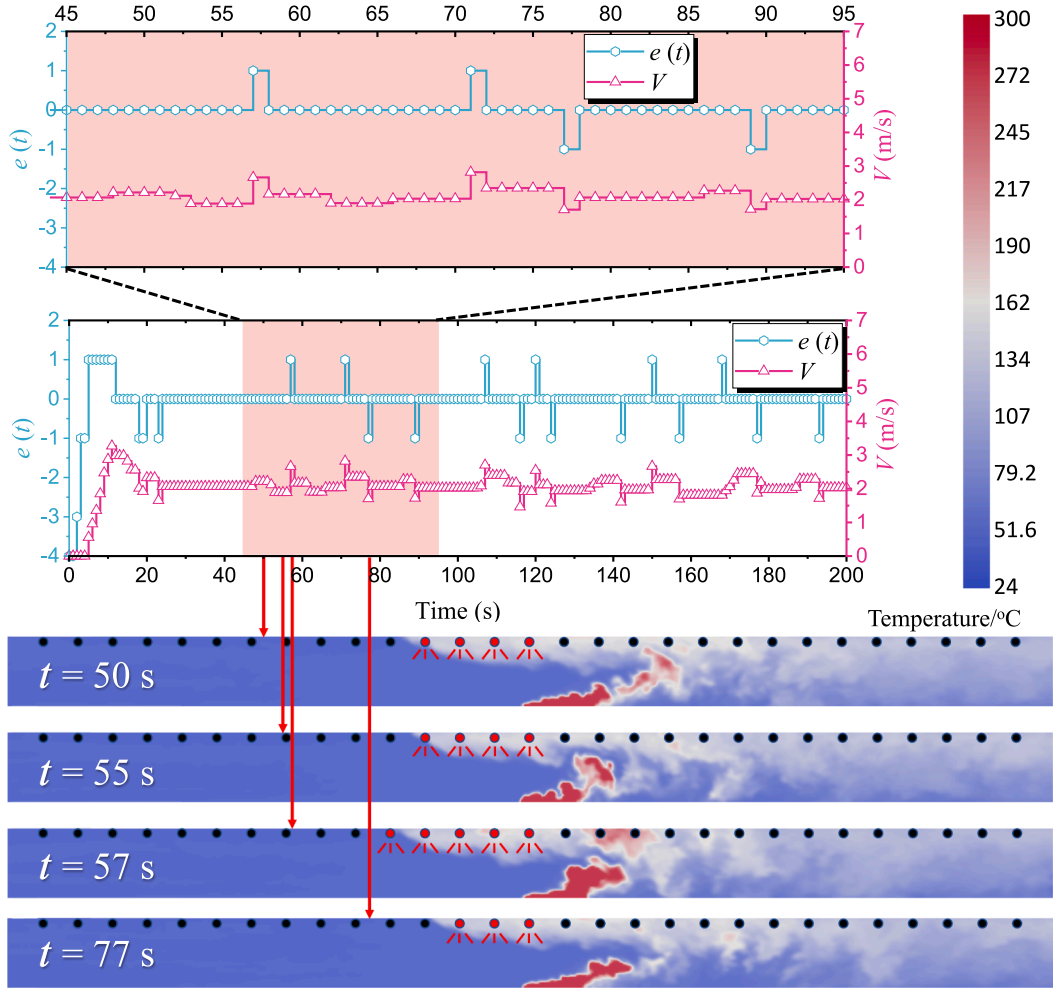


Fig. 10. Evolutions of $e(t)$ and ventilation velocity for the case that fire size is 3000 kW and $N_t = 4$, as well as several transient temperature contours. It should be noted that the data recording interval in the CFD calculation process is 1 s, so the data between each second is not displayed in the figure.

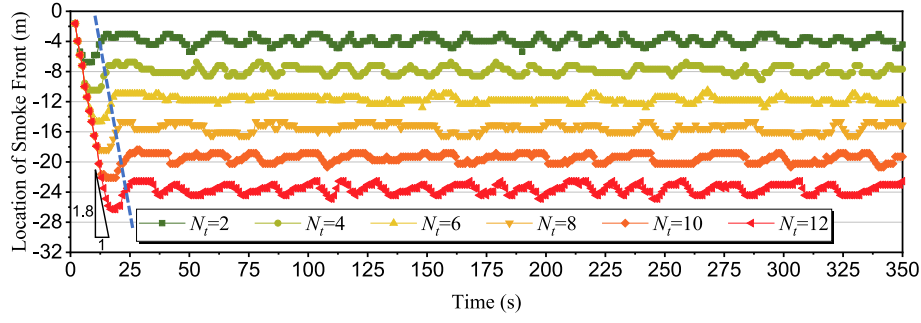


Fig. 11. Evolutions of the smoke front for different distances between the fire source and the key zone.

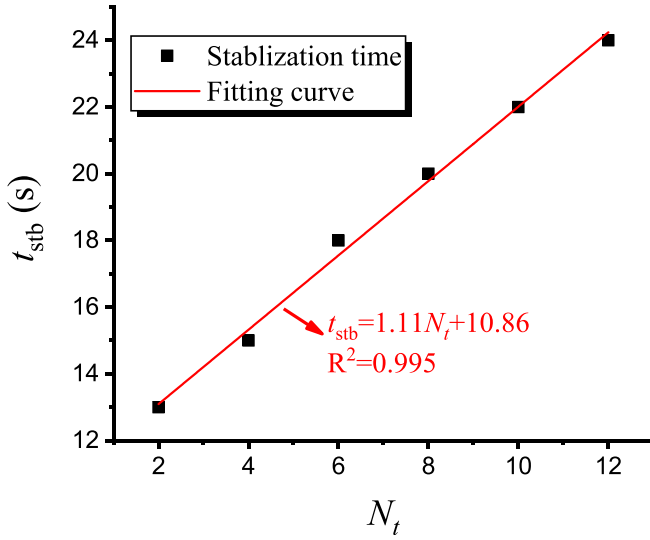


Fig. 12. Stabilization time vs. the number of the sensors in the free zone.

12 m. It is seen that the ventilation velocity almost reaches the quasi-steady state after 20 s. To ensure that the velocity reaches quasi-steady state in all cases, the average window is taken from 100 s to the maximum simulation time. As a result, the time-averaged ventilation velocity is written as

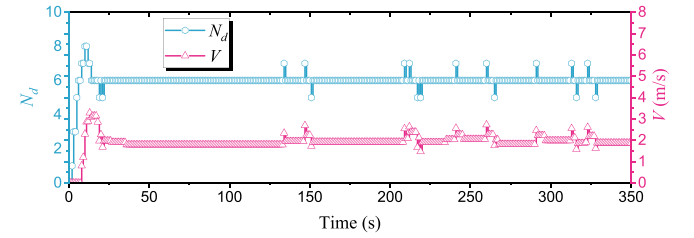


Fig. 14. Typical evolutions of $N_d(t)$ and $V(t)$. The distance between the fire source and the key zone is 14 m which responds to $N_d(t) = 6$ and 12 m back-layering length.

$$\bar{V} = \frac{\int_{100}^{t_{\max}} V(t) dt}{(t_{\max} - 100)}. \quad (12)$$

To maintain a specific back-layering length under the longitudinal ventilation, the fixed ventilation system and the proposed ventilation system should give a similar ventilation flux, though the oscillation of the ventilation velocity may cause some differences (Hong et al., 2023). Nevertheless, the velocity oscillation is small, so the comparison is still meaningful for validating the reliability of the simulations. Fig. 15 shows the comparison between the $\tilde{V}$ and $\bar{V}$ under different back-layering length and different HRRs, as the cases discussed in Section 5.2 and Section 5.3. The result shows good agreement between our simulation and the data from empirical correlation, which validates the analysis.

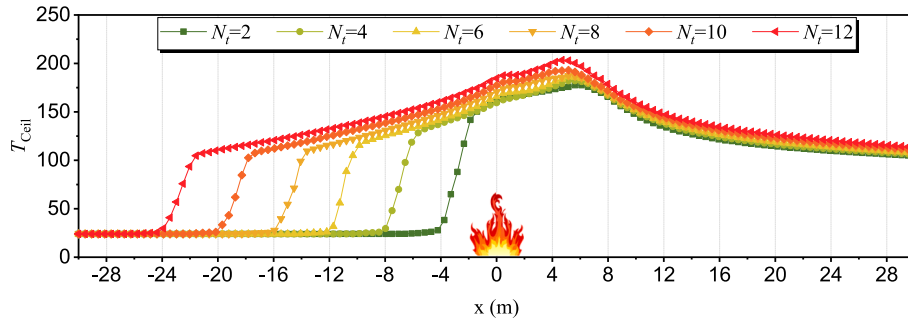


Fig. 13. Temperature distributions for different distances between the fire source and the key zone.

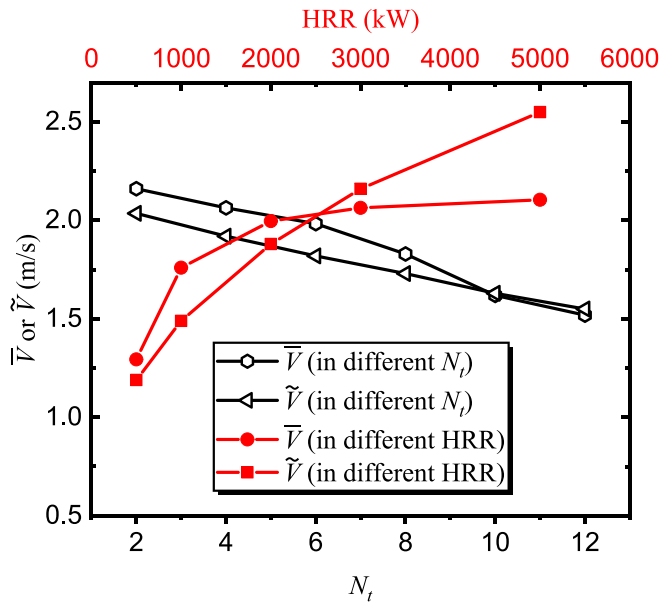


Fig. 15. The comparison between $\tilde{V}$ (Eq. (11), for fixed ventilation velocity) and $\bar{V}$ (obtained with the dynamic approach as described in the paper).

6. Conclusions

In this paper, a novel smoke control method was proposed for the protection of a key zone in a straight horizontal tunnel in case of fire. This method provides a self-adaptive ventilation scheme that can identify the fire source automatically and adjust the ventilation velocity in real time according to the information detected by sensor networks. An in-house code, combined with the CFD source code of FDS, was developed to investigate the feasibility of the proposed method numerically. The core algorithm of the self-adaptive ventilation scheme is a PID automatic controller. Its parameters significantly impact the control performance, so the PID controller parameters were determined via the magnitude analysis and the trial-and-error method.

Two types of sensors, heat and smoke detectors, were used to form the sensor networks. The numerical results show that the smoke detector can detect smoke in a broader range of HRR. The upstream smoke front is finally controlled in the tolerable zone rather than the key and free zones, which satisfies the original design concept. When the smoke is controlled by the heat detector and smoke detector networks, the time-averaged temperature distributions and the smoke mass fraction distributions beneath the tunnel ceiling are similar. It was mentioned that the proposed methodology is not directly applicable in practice with state-of-the-art point detectors, where 'old' smoke is not purged so that, once activated, the detector cannot be expected to de-activate (whereas this is an implicit assumption in the methodology as adopted here). These practical aspects are considered important, but are left for future work at this stage.

The smoke control performance for different distances between the fire source and the key zone was investigated. Using the smoke detector network, the simulation results show that the smoke can be well controlled at the desired positions (tolerable zone) for the different distances between the fire source and the key zone. Both the start-up time of ventilation and the stabilization time are approximately proportional to the distances between the fire source and the key zone. With the increased distance between the fire source and the key zone, the temperature beneath the tunnel ceiling also increases for the same fire size.

To validate the accuracy and reliability of the result in this work, the time-averaged ventilation velocity produced by the real-time ventilation system was compared to the ventilation velocity with a constant

ventilation system. The results show that the simulation results agree well with the experimental results.

CRediT authorship contribution statement

Yao Hong: Conceptualization, Investigation, Methodology, Writing – original draft, Formal analysis. **Ceji Fu:** Writing – review & editing, Funding acquisition. **Bart Merci:** Supervision, Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available upon reasonable request.

Acknowledgements

The work was supported by the China Scholarship Council (Grant No. 202106010165) and the National Key Research and Development Program of China (Grant No. 2023YFC3009102). Thanks to the computing resources, the simulation was carried out on the HPC-UGent.

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